



Google Summer of Code

DIPY

Google Summer of Code 2026 Proposal

Adding New Similarity Metrics for DIPY's Image Registration Frameworks

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Abstract

This proposal aims to extend DIPY's image registration framework by implementing new similarity metrics, with a particular focus on Mutual Information (MI) for nonrigid registration. While MI is currently available in DIPY for affine registration, support for nonrigid registration is not yet implemented despite its importance for multimodal medical imaging. This project will design, implement, and integrate an efficient MI-based similarity metric within DIPY's existing registration pipeline, together with appropriate validation and testing.

1. Organization Overview

DIPY [7] is a free and open-source Python library for the analysis of multi-dimensional medical imaging data, particularly diffusion MRI. The library provides a comprehensive collection of algorithms and tools for image processing, reconstruction, statistical analysis, machine learning, and visualization of medical images, as well as specialized methods for computational neuroanatomy and tractography.

DIPY is widely used within the medical imaging and computational neuroanatomy communities and has an active open-source development ecosystem. The project has participated in the Google Summer of Code program multiple times, fostering contributions from new developers and expanding the capabilities of the library.

2. Problem Statement

The project is framed within the context of volumetric image registration in medical imaging. Image registration aims to align multiple images taken at different times, from different viewpoints, or using different modalities [17].

It is a fundamental step in image analysis workflows where useful information is obtained by combining multiple data sources. Typical applications include multimodal image fusion, longitudinal analysis of disease progression and change detection.



Figure 1: Image from MATLAB documentation [9].

2.1. Feature-Based Registration

Classical approaches to image registration often rely on the detection and matching of corresponding features between images, such as edges, corners, or anatomical landmarks, and estimate the transformation that best aligns the detected features.

These methods present several limitations. In some cases, they require manually identified landmarks, introducing subjectivity and limiting the scalability. Automatic feature detection may be used, but the reliability of detected features can vary significantly depending on the image modality, noise level, and anatomical variability.

For these reasons, many modern medical image registration frameworks, including DIPY, rely on intensity-based approaches.

2.2. Intensity-based Image Registration

Intensity-based image registration usually formulates the alignment problem as an optimization task. Given two images I and J , the goal is to find a spatial transformation T that aligns them by optimizing a similarity metric $S(I, T(J))$. The optimization procedure searches for the transformation parameters that maximize or minimize this measure (depending on the metric used).

In the simpler case of affine registration, the spatial transformation can be expressed as

$$x' = Ax + b$$

where $A \in \mathbb{R}^{3 \times 3}$ represents the linear transformation and $b \in \mathbb{R}^3$ represents translation. This model contains only twelve parameters and can represent global geometric transformations such as translation, rotation, or scaling. Because of the small parameter space, optimization algorithms can efficiently estimate the transformation.

In contrast, nonrigid (or deformable) registration aims to estimate a dense displacement field that allows local deformations between images [13]. In this case, instead of estimating a global affine transformation, the goal is to estimate a spatial deformation field $u : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that defines a local displacement vector for every voxel in the image domain. The resulting transformation can be written as

$$T(x) = x + u(x),$$

where $u(x)$ represents the displacement applied to the voxel located at position x .

While much more expressive, this increased flexibility comes at the cost of a dramatically larger parameter space, leading to a significantly more challenging optimization process.

2.3. Similarity Metrics and Multimodal Registration

A key component of intensity-based registration is the similarity metric used to evaluate alignment between images. Common metrics include mean squared error, cross-correlation, and mutual information, which will be the main focus of this project.

Mutual Information (MI) is particularly important in the case of multimodal image registration where a direct intensity correspondence between images may not exist. Instead, it measures statistical dependence between the intensity distributions of two images, making it a well suited metric for aligning images obtained using different modalities [12].

Let I and J denote two images. The entropy of an image I , which measures the uncertainty of its intensity distribution, is defined as

$$H(I) = - \sum_i p_I(i) \log p_I(i),$$

where $p_I(i)$ represents the probability of intensity value i in image I . Similarly, the joint entropy of two images is defined as

$$H(I, J) = - \sum_{i,j} p_{IJ}(i, j) \log p_{IJ}(i, j),$$

where $p_{IJ}(i, j)$ is the joint probability distribution of the intensity pairs in both images.

The mutual information between the two images can then be written as

$$MI(I, J) = H(I) + H(J) - H(I, J).$$

In the context of image registration, the objective is to estimate a spatial transformation T that maximizes the mutual information between a reference image I and a transformed moving image $J \circ T$:

$$T^* = \arg \max_T MI(I, J \circ T). \quad (1)$$

Mutual information can be interpreted intuitively as a measure of how much knowing the intensity of a voxel in one image reduces the uncertainty about the intensity of the corresponding voxel in the other image. If the two images are statistically independent, their joint entropy is the sum of their individual entropies and the mutual information is zero. On the other hand, when the images are well aligned, the statistical dependency between their intensity distributions increases, leading to higher mutual information values.

In practice, the probability distributions required to compute mutual information are not known

beforehand and must be estimated from the image data. This is typically done using histograms of voxel intensities. For each image, a histogram is computed to estimate the distribution of its intensity values. In addition, a joint histogram is constructed by counting how often pairs of intensities from the two images occur at corresponding voxel locations.

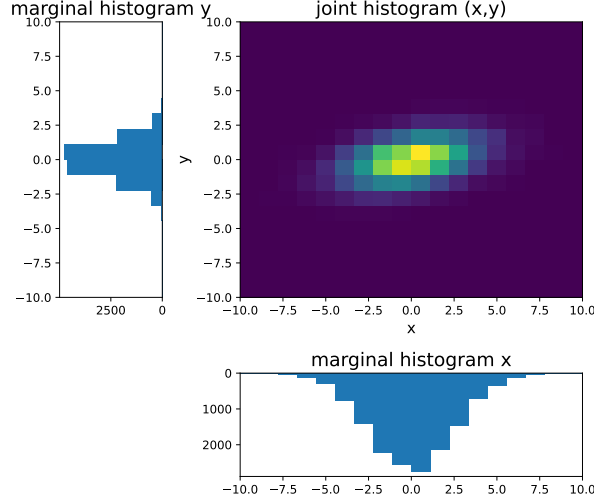


Figure 2: Joint histogram and marginal histograms illustrating the estimation of intensity distributions used to compute mutual information.

The joint histogram provides an estimate of the joint probability distribution $p_{IJ}(i, j)$, while the individual histograms provide the marginal distributions $p_I(i)$ and $p_J(j)$. These estimated distributions are then used to compute the mutual information between the two images, which serves as the objective function during the registration process.

3. Existing Infrastructure

3.1. MI for Affine Registration

Before going into the project planning in detail, this subsection provides a brief overview of the current implementation available in the repository. DIPY already implements affine image registration using a multi-resolution optimization framework that maximizes the Mutual Information (MI) between a static and a moving image. The algorithm progressively refines an affine transformation matrix $A \in \mathbb{R}^{4 \times 4}$ that defines the spatial deformation between the two images.

The current implementation of MI is located at `dipy/align/imaffine.py` and is largely based on the method proposed in [10]. The metric requires specifying the number of bins used to estimate the intensity histograms of the images. Optionally, the algorithm can also use sparse sampling, controlled by a parameter in the range $(0, 1]$, which specifies the proportion of voxels used for estimating these histograms. This allows reducing computational cost at the expense of a less accurate estimate of the intensity distributions.

The method offers a flexible interface and does not require the input images to share the same voxel spacing. Instead, the implementation internally constructs matrices that convert between voxel coordinates and physical coordinates (in millimeters). The affine transformation is estimated in the physical coordinate system. During gradient computation, the voxel spacing is

also used to correctly scale derivatives along each spatial direction. Additionally, the implementation accepts optional masks so that histogram statistics are computed only inside relevant regions of the images.

The estimation of the joint intensity distribution is implemented in Cython in `dipy/align/parzenhist.pyx`. This module follows the Parzen window approach described in [10], where the joint intensity distribution of the static and moving images is approximated using smooth kernel functions rather than a discrete histogram.

Using Parzen window estimation provides a differentiable approximation of the joint intensity distribution, which allows the mutual information metric to be optimized with gradient-based methods. The implementation is written in Cython for efficiency, enabling the evaluation of image samples and the corresponding transformation Jacobians inside `nogil` loops.

The module therefore provides the core routines required to estimate the joint histogram, compute the mutual information value, and accumulate the derivatives needed by the optimization procedure during image registration.

3.2. Nonrigid Registration

DIPY also provides a framework for nonrigid image registration implemented in `dipy/align/imwarp.py`. In particular, this module contains the implementation of diffeomorphic registration algorithms such as the Symmetric Diffeomorphic Registration (SyN) method.

Unlike the affine registration framework, mutual information is not currently available as a similarity metric for nonrigid registration. Instead, the existing implementation relies on alternative metrics including normalized cross-correlation, expectation-maximization, and the sum of squared differences. These metrics are implemented in `dipy/align/metrics.py`.

The design of the registration framework is largely agnostic to the specific similarity metric being used. The optimizer interacts with the metric through the abstract interface defined by the `SimilarityMetric` class, which specifies the methods required to evaluate the similarity measure and compute the update fields used during optimization. As a result, new similarity metrics can be incorporated into the framework by implementing this interface.

For consistency with the existing codebase, any new metric implementation should follow the same interface and design principles. The currently available implementations provide useful references, as they illustrate how the gradient of the similarity measure with respect to the deformation field is computed and integrated into the optimization procedure.

The existing metrics are grounded in well-established methods in the registration literature. Cross-correlation follows the formulation proposed in [2], the expectation-maximization metric is based on the approach described in [1], and the sum of squared differences implementation follows the diffeomorphic demons framework [4].

4. Implementation Plan

Although DIPY already provides an implementation of Mutual Information (MI) for affine registration, this implementation cannot be directly reused within the nonrigid registration framework. The affine registration module optimizes a low-dimensional vector of transfor-

mation parameters, whereas the nonrigid framework operates over dense displacement fields defined across the image domain. As a result, the existing `MutualInformationMetric` cannot be directly integrated into the nonrigid optimization pipeline.

Supporting MI within the nonrigid framework therefore requires the implementation of a new similarity metric compatible with the `SimilarityMetric` interface used by the diffeomorphic registration algorithms. In particular, the metric must provide the methods `compute_forward` and `compute_backward`, which produce voxelwise update fields used by the optimizer to iteratively refine the deformation field.

While the optimization interface differs from the affine case, parts of the existing MI infrastructure can still be reused. In particular, the `ParzenJointHistogram` implementation already provides efficient routines for estimating the joint intensity distribution using Parzen windows. This smooth histogram formulation enables the computation of the mutual information value and its derivatives, which can be incorporated into the update fields required by the nonrigid optimizer.

The proposed implementation will therefore build on the existing histogram estimation routines while adapting the gradient computation to operate with respect to the displacement field used in the nonrigid registration framework.

4.1. Literature Review

The initial phase of the project focuses on studying the relevant literature on mutual information and its application to nonrigid image registration. This work began during the proposal preparation period and will continue during the first weeks of the project.

In particular, the works of [14] and [5] explore statistical similarity measures for multimodal nonrigid registration, including mutual information. These papers describe variational formulations in which the joint intensity distribution is estimated using smooth Parzen window density estimators, allowing the similarity measure to be differentiated with respect to the deformation field.

Although the precise implementation details will require further study, these formulations appear to provide a useful theoretical reference for developing a mutual information metric compatible with the diffeomorphic registration framework used in DIPY.

In addition to the academic literature, existing implementations in other registration frameworks were also examined. In particular, the Insight Toolkit (ITK) [16] provides a working implementation of mutual information metrics that can be used together with dense transformation models. The source code is publicly available and includes both the classical mutual information metric and Parzen window-based formulations for estimating the joint intensity distribution.

Although the architecture of ITK differs significantly from that of DIPY, studying these implementations may provide useful insight into practical aspects of computing the mutual information value and its derivatives during registration.

Finally, the original project description mentions that additional reference implementations not currently included in DIPY may be provided to the selected contributor.

4.2. Metric Implementation

Before beginning the implementation, the mathematical formulation of the algorithm will be carefully reviewed and translated into a high-level pseudocode description. This step will help clarify how the mutual information formulation must interact with the nonrigid registration framework and which parts of the existing infrastructure can be reused.

Special attention will be given to the efficiency of the implementation. Operations that are expected to be computationally intensive may need to be implemented in Cython and executed inside `nogil` sections, following the strategy already used in the existing Parzen histogram implementation.

Once the design and pseudocode have been validated by the mentors, the coding stage will begin. The implementation will follow the structure of the existing similarity metrics in `dipy/align/metrics.py`, ensuring compatibility with the `SimilarityMetric` interface and consistency with the coding practices used throughout the DIPY codebase.

4.3. Testing and Validation

To ensure correctness and robustness, the implementation shall be validated following the testing practices followed in DIPY. Existing unit tests for similarity metrics and Parzen histogram estimators could be used as reference when designing new tests for the mutual information metric.

New tests will be introduced if needed to verify that the metric correctly implements the `SimilarityMetric` interface and produces valid forward and backward update fields for both 2D and 3D cases. If possible, the behavior of the implementation could be compared against simplified reference computations or approximations to verify that the computed updates are consistent with the expected gradient of the similarity measure. A similar procedure is followed for other metrics.

Additionally, integration tests may be added to confirm that the metric interacts as expected with the optimizer and produces meaningful registrations.

Finally, some benchmarking experiments may be carried out in comparison with existing similarity metrics in order to evaluate the impact of the proposed implementation. Both quantitative and qualitative comparisons can be used to assess the behavior of the new metric within the nonrigid registration framework. The original project description suggests that an improvement in registration accuracy may be expected when using mutual information in multimodal scenarios. However, the observed performance may depend on the characteristics of the images used as well as on the evaluation criteria adopted for the comparison.

4.4. Project Timeline

The timeline below is preliminary and may be adjusted based on mentor feedback and project progress. The proposed work corresponds to a 90-hour project spanning eight weeks. As discussed in Section 7, the implementation may later be extended beyond the initial objectives outside the scope of GSoC 2026.

Weeks 1–2 – Literature Review and Implementation Design

- Review relevant literature on mutual information for nonrigid registration.
- Study existing implementations in DIPY and reference frameworks such as ITK.
- Develop a pseudocode description of the algorithm and identify reusable components within the current codebase.
- Define the structure of the new metric following the `SimilarityMetric` interface.
- Design how the mutual information metric will interact with the nonrigid optimizer.
- Identify potential computational bottlenecks and determine whether parts of the implementation should be moved to Cython.

Deliverables:

- A clear algorithmic description and pseudocode of the proposed implementation, and a preliminary design outlining how the metric integrates with the existing registration framework.

Weeks 3–6 – Metric Implementation and Integration

- Implement the mutual information metric for nonrigid registration following the design produced in the previous phase.
- Reuse the existing Parzen histogram infrastructure where possible.
- Integrate the metric into the nonrigid registration pipeline during development.
- Move computationally intensive components to Cython if necessary.

Deliverables:

- A working implementation of the mutual information metric compatible with the `SimilarityMetric` interface integrated within the nonrigid registration framework.

Week 7 – Testing

- Implement unit tests following the patterns used in existing DIPY metrics.
- Validate the update fields and metric behavior on synthetic examples.
- Ensure compatibility with both 2D and 3D cases where applicable.

Deliverables:

- A comprehensive test suite covering the new metric.

Week 8 – Validation and Documentation

- Perform qualitative and quantitative benchmarking against existing metrics.
- Evaluate the behavior of the new metric on representative registration tasks.
- Write documentation and finalize the code for submission.

Deliverables:

- Benchmarking results comparing the new metric with existing alternatives.
- Documentation describing the metric, its usage, and implementation details.
- Finalized code ready for review and integration into the DIPY codebase.

While the timeline above is structured into distinct phases, in practice some activities may overlap. In particular, the implementation phase will likely begin during the design stage and continue throughout the project as new insights are obtained, as is common during scientific software development.

5. Potential Project Extensions

So far, the proposal has focused on the core objectives of the project, following the expectations outlined in the original DIPY project description while keeping the scope realistic and avoiding unnecessary overextension. However, I have identified several additional ideas that could potentially be valuable for the library. In this section, I briefly outline a set of possible extensions, presented without a particular order of priority, which could be discussed with the project mentors. If time allows, some of these ideas could be explored during the GSoC period. Otherwise, they may serve as potential follow-up contributions after the completion of the main project.

5.1. Adaptive Histogram Binning

The current mutual information implementation requires the user to manually specify the number of bins used to estimate the intensity histograms, falling back to a default value of 32. While this parameter offers flexibility, selecting an appropriate value can be nontrivial and may influence the accuracy of the metric. As a possible extension, adaptive strategies for automatically determining the number of histogram bins could be explored. Statistical rules such as the Freedman-Diaconis rule [6] or the Scott’s rule [11] could be used to estimate a suitable bin width based on the intensity distribution of the images. Incorporating such heuristics into the metric could reduce the need for manual parameter tuning while maintaining compatibility with the existing implementation.

5.2. Benchmarking and Reproducible Evaluation

In the original project proposal, it is stated that the introduction of a Mutual Information (MI) metric for nonrigid registration is expected to improve registration accuracy, particularly in multimodal scenarios. However, in order to provide a fair and meaningful comparison across

different similarity metrics and registration methods, a well-defined benchmarking framework is required. At the moment, the benchmarking infrastructure available in `dipy/benchmarks` appears to be relatively limited.

Developing a more comprehensive benchmarking setup for evaluating registration techniques across different datasets and modalities, in terms of both accuracy and computational performance, could therefore be a valuable extension. Such a framework would not only provide a solid basis for comparing the proposed metric with existing methods, but could also serve as a reference for evaluating future additions to the codebase. In this way, it could help establish clearer evaluation practices and provide a useful template for benchmarking other algorithms implemented in DIPY.

Several quantitative measures could be used to assess performance. When anatomical labels are available, overlap-based metrics such as the Dice similarity coefficient could be used to measure the agreement between the segmentations. In addition, landmark-based metrics such as target registration error could be employed when suitable landmarks are available. Computational performance could also be evaluated by measuring runtime and memory usage during the optimization process.

These experiments could be provided as reproducible scripts or Jupyter notebooks illustrating how the different similarity metrics behave on representative datasets, serving both as validation tools and as educational resources.

5.3. Synthetic Deformation

Inspired by issue [#1687](#), another possible extension would be the development of a synthetic framework for evaluating registration algorithms under controlled conditions. In this approach, a known random deformation field would be applied to a reference image in order to generate artificially warped images. Since the ground-truth deformation is known, the performance of a registration method can be evaluated directly by measuring how accurately the algorithm recovers the applied transformation. Such a tool could be useful not only for benchmarking and validating registration methods, but also for introducing additional tests under `dipy/align/tests`, helping ensure that algorithms behave correctly and maintain their expected performance as the codebase evolves.

6. On the use of AI

I agree with DIPY's position regarding the responsible use of AI tools in the development process. AI-assisted tools may be used as supporting instruments to improve productivity, just as they were used during the preparation for this proposal (for improving clarity and correcting language issues). However, their role is strictly supportive. I commit to ensuring that all code, algorithms, and documentation produced during the project will be carefully reviewed and fully understood. Ultimately, I take full responsibility for the final implementation and its correctness.

7. Future Work

My goal for this project is for it to serve as my introduction to the world of open-source contribution. In particular, I aim to become a regular contributor to DIPY. The library is already used in my current workplace (see Section 8), and contributing to its development would therefore be both professionally and personally valuable.

Through this project, I expect to gain familiarity with the codebase and with the development practices followed by the project. Beyond the scope of this proposal, I hope to continue contributing to the library by improving and extending existing functionality, helping with issues, and supporting ongoing development where useful.

In addition, I have a few preliminary ideas for potential extensions and future work that may or may not align with DIPY’s roadmap. These are briefly outlined in this section.

7.1. Implementing Structural Similarity Index (SSIM) as a new registration metric

One potential extension is the integration of the Structural Similarity Index (SSIM) as an alternative similarity metric for nonrigid registration. Unlike traditional intensity-based measures such as SSD or mutual information, SSIM evaluates similarity using local statistics, including mean intensity, variance, and cross-covariance computed over image patches. This formulation allows the metric to emphasize structural consistency between images rather than relying solely on absolute intensity differences.

As mentioned in Section 8, my current work involves the use of generative artificial intelligence for medical image registration, where SSIM is commonly used both as a training objective and as a quantitative evaluation metric. Exploring its integration into the existing DIPY registration framework could therefore provide a useful complementary similarity measure alongside the metrics already available in the library.

7.2. Neural Parameterization of Deformation Fields

One potential research direction that may be explored beyond the scope of this project concerns alternative parameterizations of the deformation field used in diffeomorphic registration.

In classical approaches, the displacement field $u(x)$ is directly optimized over the image domain. While effective, this representation may involve a large number of parameters and can lead to complex optimization landscapes. An alternative strategy is to represent the deformation field using a parametric function approximator, such as a small neural network $f_\theta(x)$, and optimize its parameters instead of the displacement values at each voxel.

In this formulation, the neural network acts as a continuous model of the deformation field,

$$u(x) = f_\theta(x),$$

and the parameters θ can be optimized through backpropagation using the same similarity metrics employed in classical registration frameworks.

Related ideas have recently been explored in the literature through implicit neural representations and neural-field models for deformable image registration [15]. While such approaches

differ from traditional deep learning pipelines that rely on large training datasets, they offer an interesting alternative parameterization of deformation fields and may provide advantages in terms of regularization and representation efficiency.

This direction is only a preliminary idea and would require further study to determine whether it aligns with the design philosophy and scope of the DIPY library.

7.3. Learning-Based Registration

Another possible direction for future work would be to explore the integration of lightweight pretrained deep learning models for approximate image registration. In recent years, several learning-based approaches [3, 8] have demonstrated the ability to predict deformation fields directly from pairs of images, often providing significantly faster inference compared to classical iterative optimization methods.

DIPY could provide a mechanism for loading lightweight pretrained models that predict deformation fields for specific types of data. These models would not replace the existing variational registration algorithms, but could instead serve as fast initial approximations or alternative registration strategies in scenarios where speed is important, such as large-scale or batch processing workflows.

Such models could be distributed separately from the main library and loaded through a lightweight interface, preserving the core design philosophy of DIPY while allowing users to experiment with learning-based registration approaches. When compatible hardware is available, this could offer a practical way to accelerate certain registration tasks while maintaining interoperability with the existing DIPY registration framework.

8. Applicant Information

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Education

- **M.Sc. in Artificial Intelligence** — Universidad Politécnica de Madrid (2024–2025)
- **Double Bachelor’s Degree in Mathematics and Computer Science** — Universidad de Sevilla (2019–2024)

Research and Professional Experience

- **Computer Vision Researcher**, Medical Imaging Lab PROMISE-LAIMBIO, Universidad Rey Juan Carlos (October 2025–Present).

Research focused on computer vision and medical imaging as part of the project MAGERIT (Medical AuGmentEd Reality and DIgital Twins), a collaborative research initiative aimed at developing patient-specific digital twins. My current work focuses on medical image registration using generative artificial intelligence, modeling plausible anatomical deformations through generative models.

- **AI Researcher**, ML Circle, Madrid (January 2025–October 2025).

Research and technological consultancy on generative font shape modeling as part of my Master’s thesis. The source code for this project will soon be publicly available, as the work is has been recently accepted in the *Computers & Graphics* journal.

Previous Contributions

- Open pull request for DIPY implementing a Cython optimization for the nested Python triple loop used in the `localpca eig` path. [#3806](#).

Availability

I am currently working as a full-time researcher. However, the proposed project is closely related to my current work, which makes it feasible to allocate dedicated time to it. With the approval of my supervisor and thanks to the flexible nature of my research schedule, I will be able to dedicate the necessary time to successfully complete the project. On average, I expect to be able to dedicate approximately two and a half hours per day from Monday to Friday, with additional time available during weekends if needed, as well as occasional fully dedicated days when approaching key milestones or deliverables in the project timeline.

At the moment, I do not have any planned commitments during June or July that would interfere with the proposed timeline. Should unforeseen circumstances arise, I would still be able to adjust my schedule accordingly due to the flexibility of my working environment.

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